STAT 220B Cheatsheet: The Story of the Course

Decision Theory through Lecture 11

This cheatsheet traces the **logical flow** of the course. For each concept, three questions: **why** is it needed, **how** does it connect to what came before, and **when** do you reach for it. It is a map, not a substitute for the lectures.

The five ingredients of every decision problem

Every problem from Lecture 1 onward has the same five components.

Symbol	**Name**	**What it is**
Θ	Parameter space	Set of possible truths θ
$\mathcal{A}$	Action space	Set of possible decisions
$L(\theta, a)$	Loss function	Cost of action a when truth is θ. Bounded below: $L \geq -K$
$X \in \mathcal{X}$, $X \sim P_\theta$	Data and model	Observations whose distribution depends on θ
$\pi(\theta)$	Prior (optional)	Distribution on Θ encoding belief or weighting

A **decision rule** $\delta : \mathcal{X} \to \mathcal{A}$ takes data to actions. A **randomized rule** $\delta^*(x, \cdot)$ is a distribution on $\mathcal{A}$ for each x. In no-data problems, a rule is just an action.

Two viewpoints for evaluating a rule

The most important table in the course. Every notation confusion traces back to forgetting which side you are on.

	Frequentist viewpoint	**Bayesian viewpoint**
Average over	Data X, holding θ *fixed*	Parameter θ, possibly given X
Main object	$R(\theta, \delta) = \mathbb{E}_{P_\theta}[L(\theta, \delta(X))]$ *risk function*	$\rho(\pi, a) = \mathbb{E}^\pi[L(\theta, a)]$ *Bayesian expected loss*
Returns	A *function* of θ	A *number* for each action
Compare rules by	Partial order: δ_1 R-better than δ_2 iff $R(\theta, \delta_1) \leq R(\theta, \delta_2)$ for all θ, strict for some	Direct comparison: pick the smallest
Bridge via prior	$r(\pi, \delta) = \mathbb{E}^\pi[R(\theta, \delta)]$, the *Bayes risk* of δ	Average posterior expected loss over the marginal $m(x)$ of X

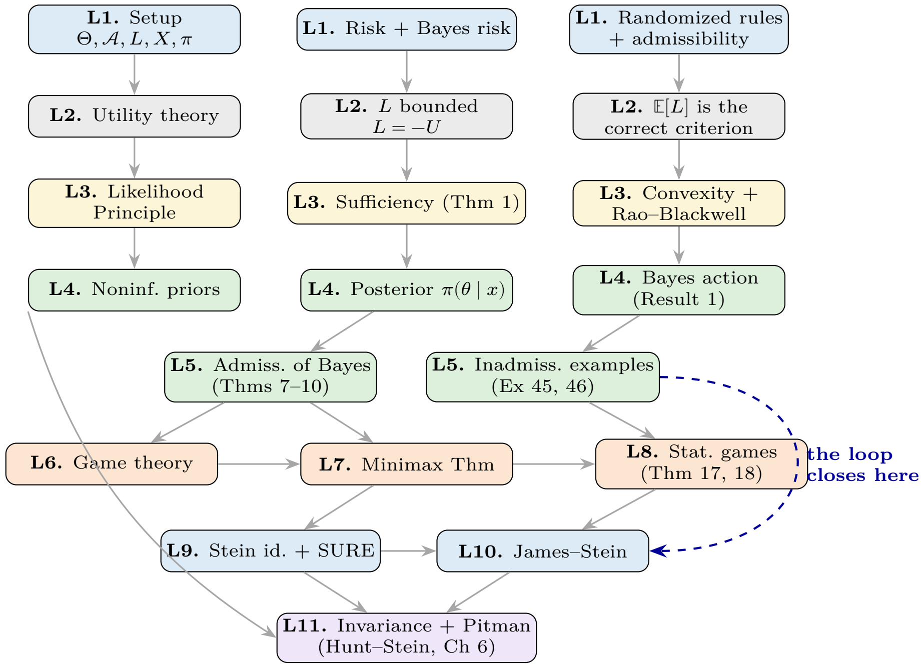
Notation tag (the thing that trips everyone). $\mathbb{E}_{P_\theta}[\cdot]$ averages over the data with θ fixed. $\mathbb{E}^\pi[\cdot]$ averages over θ with prior or posterior π. $\mathbb{E}[\cdot \mid T = t]$ is conditional over data given a sufficient statistic. **Test:** ask "what is random in this expression, and what is fixed?''

The crucial identity (Result 1 of §4.4, used in every Bayes calculation):

$$r(\pi, \delta) \;=\; \mathbb{E}^\pi[R(\theta, \delta)] \;=\; \mathbb{E}^{m(x)}\big[\rho(\pi(\theta \mid x),\, \delta(x))\big].$$

A Bayes rule can be found by minimizing posterior expected loss pointwise in x ("extensive form''). This is what you use in practice.

The course as a flow diagram



Color key. Blue framework and Stein. Gray foundations (utility). Yellow structural tools. Green Bayesian / admissibility. Orange minimax. Purple invariance. The **dashed arrow** is the punchline: Example 46 in Lecture 5 flagged that the sample mean is inadmissible for $p \geq 3$. Lectures 9–10 close the loop by proving it. Lecture 11 synthesizes: invariance gives a structural reason for the original minimax claim but does not protect against the inadmissibility.

Narrative: why each layer exists

Layer 1: Setup (Lecture 1). The five ingredients, the risk function $R(\theta, \delta)$ and Bayes risk $r(\pi, \delta)$, admissibility, randomization, and three selection principles (Bayes, minimax, invariance). *Why needed:* Statistics is not just estimation. It is deciding under uncertainty, with consequences. We need a framework that handles the cost of being wrong, not just the standard error. *Unsolved problem at end:* risk functions cross, so admissibility alone does not pick a unique rule. We need selection principles, plus tools to simplify the search.

Layer 2: Utility theory, the foundation under everything (Lecture 2). *Why it exists (and why it did not feel obvious in lecture).* Lecture 1 takes $L(\theta, a)$ as a given input and announces that we will minimize $\mathbb{E}[L]$. Two questions are left dangling: (1) Where does L come from? (2) Why is minimizing $\mathbb{E}[L]$ the right thing to do, rather than, say, the median loss, the worst-case loss, or some other functional? Utility theory answers both, and provides a property of L that the next four layers all silently use.

The motivating puzzle. Offered (A) \$200,000 for certain, or (B) a 50–50 chance of \$0 or \$410,000, most people pick (A) even though (B) has higher expected dollars. "Value'' is not linear in money. We need a numerical encoding of preferences that captures this. That is what a utility function U does.

The axioms (von Neumann–Morgenstern). Preferences over distributions of consequences satisfy: (1) completeness, (2) transitivity, (3) independence (mixing with a third distribution preserves preference), (4) continuity (no consequence is infinitely good or bad). Under these axioms:

1. There exists a utility function $U : \mathcal{R} \to \mathbb{R}$ such that $P_1 \prec P_2 \iff \mathbb{E}^{P_1}[U] < \mathbb{E}^{P_2}[U]$. Unique up to positive affine transformation.

2. U is **bounded** (forced by Axiom 4; the St. Petersburg paradox is the demonstration).

3. Define $L(\theta, a) = -U(\theta, a)$. Then maximizing $\mathbb{E}[U]$ is the same as minimizing $\mathbb{E}[L]$.

Why utility matters more than it looks. Three downstream consequences, each invoked later without explicit reference back here.

- It **justifies expected loss** as the criterion. Minimizing $\mathbb{E}[L]$ is not one option among many. It follows from the axioms.

- It **forces L to be bounded below by $-K$**, which is exactly the assumption that makes the admissibility theorems of Lecture 5 work (Theorems 7–9 all need finite Bayes risk, which needs L bounded).
- It **resolves the St. Petersburg paradox**: naive expected monetary value gives an infinite price to play, but bounded utility removes the pathology. The same fix prevents formal Bayes pathologies later.

A variant worth knowing: regret loss. $\ell(\theta, a) = \sup_{a'} U(\theta, a') - U(\theta, a)$. Penalty relative to the best possible action. For Bayesian analysis, equivalent to $L = -U$ (they differ by a function of θ alone, which does not affect minimization). Can differ for minimax, where we return to it in Chapter 5.

Short version, in one sentence. Utility is the load-bearing wall behind the framework. The rest of the course assumes its conclusions ($\mathbb{E}[L]$ is the criterion, L is bounded), uses them, and never looks back.

Layer 3: Simplifying tools (Lecture 3). The decision rule space $\mathcal{D}^*$ is enormous. Three theorems shrink the search before any optimization begins.

- **Likelihood Principle (motivating idea).** The 9-heads / 3-tails binomial vs. negative-binomial example: same likelihood, different p-values. Classical analysis can violate the LP. The LP suggests all relevant information about θ lives in the likelihood, which lives in a sufficient statistic.
- **Sufficiency (Theorem 1).** If T is sufficient for θ, every rule δ_0 is R-equivalent to some rule based on T. *Search only over rules of T.*
- **Convexity + Jensen ⇒ Theorem 3.** If L is convex in a, any randomized rule is dominated (in pointwise loss) by its mean. *Search only over nonrandomized rules.*
- **Rao–Blackwell (Theorem 4).** Combines both: $\delta_1(t) = \mathbb{E}[\delta_0(X) \mid T = t]$ is nonrandomized, based on T, and R-better than δ_0 whenever the loss is convex.

Connection upward. Theorem 3 needs L convex in a. Where did convexity become a reasonable assumption? Utility theory (Layer 2) put L in the picture as a quantitative loss; convex losses (squared error, absolute error, log loss) are then natural choices. *Connection downward:* every Bayes, minimax, and Stein rule we build below quietly uses this reduction. You never write down a randomized rule again unless an adversary is involved.

Layer 4: Bayesian decision theory (Lecture 4). The mechanics of computing optimal rules given a prior.

- **Noninformative priors** for when no prior information is available: location $\pi(\theta) = 1$, scale $\pi(\sigma) = 1/\sigma$, Bernoulli Jeffreys $\pi(p) \propto [p(1-p)]^{-1/2}$. All improper.
- **Posterior** $\pi(\theta \mid x) \propto f(x \mid \theta)\pi(\theta)$. For improper priors this is *formal*, valid if the marginal $m(x)$ is finite.
- **Posterior expected loss** $\rho(\pi(\theta \mid x), a) = \mathbb{E}^{\pi(\theta|x)}[L(\theta, a)]$ and the **Bayes action** $\delta^\pi(x) = \arg\min_a \rho(\pi(\theta \mid x), a)$.
- **Result 1** (extensive form = normal form): a rule is Bayes iff it minimizes posterior expected loss pointwise in x. This is the identity from Section 2 above, and it is the single most-used tool in the course.

Three workhorse Bayes actions, one for each common loss:

Loss	**Bayes action**	**Why**
$(\theta - a)^2$ (squared error)	Posterior mean $\mathbb{E}[\theta \mid x]$	Variance decomposition: $\arg\min \mathbb{E}[(\theta - a)^2] = \mathbb{E}[\theta]$
$\lvert\theta - a\rvert$ (absolute error)	Posterior median	Calculus on $\int \lvert t - a\rvert dF(t)$
0–1 loss	Posterior mode (MAP)	Direct minimization on indicator loss

Layer 5: Admissibility of Bayes rules (Lecture 5). The question: are Bayes rules admissible?

The good news (§4.8.1, proper priors with finite Bayes risk). Theorem 7: discrete Θ, prior charges every point ⇒ admissible. **Theorem 8: unique Bayes rule ⇒ admissible** (used most often). Theorem 9: continuous risks, prior charges every open set ⇒ admissible. *All three need finite Bayes risk, which is where bounded loss from Layer 2 (utility theory) shows up.*

The bad news (§4.8.2, improper priors / generalized Bayes rules). When $r(\pi, \delta) = \infty$, the Bayes-risk argument fails. Two cautionary examples:

- **Example 45 (gamma scale).** $\pi(\beta) = 1/\beta$ for Gamma(α, β). The generalized Bayes rule is dominated by a constant multiple, with risk ratio $(\alpha + 1)^2/(\alpha - 1)^2$.
- **Example 46 (Stein).** $\pi(\theta) = 1$ on $\mathbb{R}^p$ with $\mathbf{X} \sim \mathcal{N}_p(\theta, I_p)$. The generalized Bayes rule $\delta^0(\mathbf{x}) = \mathbf{x}$ is inadmissible for $p \geq 3$. *This is the example Lectures 9–10 will actually prove.*

Long-run motivation (§4.8.3). Even a Bayesian should care about admissibility: using an inadmissible rule repeatedly produces, with probability one, a higher long-run average loss than a dominating rule. Admissibility is not a Bayesian-vs-frequentist issue; it is a sanity check.

Layer 6: Minimax theory (Lectures 6–8). If no prior is available, choose δ to minimize $\sup_\theta R(\theta, \delta)$, protecting against the worst case.

- **L6 (game theory).** Two-person zero-sum games, randomized strategies, upper and lower values. Theorem 2 (verification by inequality); Theorem 4 (carrier of optimal strategies).
- **L7 (Minimax Theorem, Thm 15).** For finite Θ and bounded loss, the game has a value, and maximin and minimax strategies exist. Proved via S-game reformulation and separating hyperplanes (supplementary notes).
- **L8 (statistical games).** Theorem 17 (equalizer + Bayes $\Rightarrow$ minimax) and Theorem 18 (sequence of priors for improper limits).

Practical recipe. (1) Try a conjugate family π_α. (2) Compute Bayes rule δ^{π_α}. (3) Compute risk $R(\theta, \delta^{\pi_\alpha})$. (4) Solve for α that makes R *constant in* θ. (5) Conclude (Thm 17): δ^{π_α} is minimax, π_α is least favorable.

Two foundational examples to keep in mind:

Problem	**Minimax rule**	**Prior used**
$X \sim \text{Bin}(n, p)$, squared loss	$\delta(x) = (x + \sqrt{n}/2)/(n + \sqrt{n})$	$\text{Beta}(\sqrt{n}/2, \sqrt{n}/2)$ (proper; Thm 17)
$\bar{X} \sim \mathcal{N}(\theta, \sigma^2/n)$, squared loss	$\delta(\bar{x}) = \bar{x}$ (sample mean)	$\mathcal{N}(0, \tau^2)$, $\tau \to \infty$ (improper; Thm 18)

Layer 7: The Stein phenomenon (Lectures 9–10). The setup left over from Lecture 5. $\mathbf{X} \sim \mathcal{N}_p(\theta, I_p)$, squared-error loss. The sample mean $\delta^0(\mathbf{x}) = \mathbf{x}$ is MLE, UMVUE, minimax, with constant risk p. But Example 46 said it is inadmissible for $p \geq 3$. How is that possible?

- **Stein's identity (L9).** For $\mathbf{X} \sim \mathcal{N}_p(\theta, I_p)$ and smooth $\mathbf{g}$: $\mathbb{E}_\theta[(\mathbf{X} - \theta)^T \mathbf{g}(\mathbf{X})] = \mathbb{E}_\theta[\nabla \cdot \mathbf{g}(\mathbf{X})]$. Proved by integration by parts. Eliminates θ from risk computations.
- **SURE (L9).** For $\delta(\mathbf{x}) = \mathbf{x} + \mathbf{g}(\mathbf{x})$: $\hat{R}(\mathbf{x}) = p + 2\nabla \cdot \mathbf{g}(\mathbf{x}) + \|\mathbf{g}(\mathbf{x})\|^2$ is an unbiased estimator of risk. Risk dominance reduces to a pointwise comparison of SURE.
- **James–Stein (L10).** With $\mathbf{g}(\mathbf{x}) = -(p-2)\mathbf{x}/\|\mathbf{x}\|^2$: $\hat{R}^{JS}(\mathbf{x}) = p - (p-2)^2/\|\mathbf{x}\|^2$, hence $R(\theta, \delta^{JS}) = p - (p-2)^2 \mathbb{E}_\theta[1/\|\mathbf{X}\|^2] < p$ for all θ.

Why it matters beyond the proof. Stein's phenomenon is the foundation of modern shrinkage: empirical Bayes, ridge regression, lasso, hierarchical models, wavelet denoising (SureShrink). In each case, high-dimensional geometry creates statistical leverage that one-dimensional intuition does not see.

Layer 8: Invariance, the fourth selection principle (Lecture 11). If a decision problem has symmetry, restrict attention to rules that respect the symmetry. This pins down a one-parameter family without requiring a prior or worst-case computation. The fourth (and final) selection principle from Lecture 1.

The motivating idea. A particle decay problem in seconds vs. minutes must give the same physical answer. The demand of unit-blindness forces $\delta(x) = Kx$ (a one-parameter family), collapsing the infinite-dimensional rule space.

Formal setup. An invariant problem has three groups $\mathcal{G}, \bar{\mathcal{G}}, \tilde{\mathcal{G}}$ acting on $\mathcal{X}, \Theta, \mathcal{A}$, with model invariance ($X \sim P_\theta \Rightarrow gX \sim P_{\bar{g}\theta}$) and loss invariance ($L(\bar{g}\theta, \tilde{g}a) = L(\theta, a)$). A rule is **equivariant** if $\delta(gx) = \tilde{g}\delta(x)$.

Key consequences.

- **Theorem 1.** Equivariant risk is constant on orbits of $\bar{\mathcal{G}}$. For transitive groups (location, scale), the orbit is all of Θ, so equivariant rules have *constant risk in* θ. The minimax problem becomes trivial.
- **Maximal invariants (§6.3).** Analog of sufficiency for invariance. Reduces the rule space.
- **Pitman estimator (§6.4).** For location families with squared-error loss, the best equivariant rule is $\delta^*(X) = X_n - \mathbb{E}_0[X_n \mid T(X)]$, which equals the generalized Bayes rule under the uniform improper prior. For normal location, $\delta^* = \bar{X}$.
- **Hunt-Stein theorem (§6.5).** For amenable groups (location, scale, location-scale), the best equivariant rule is *minimax*. So invariance gives a structural route to minimax that does not require equalizer-Bayes arguments.

Connection to Layer 7 (the conflict). For $\mathcal{N}_p(\theta, I_p)$, the best location-equivariant estimator is $\bar{X} = \mathbf{X}$, which is minimax by Hunt-Stein. But Lecture 10 showed it is inadmissible for $p \geq 3$. The James-Stein dominator is *not* equivariant: it shrinks toward an arbitrary point, breaking translation symmetry. **Invariance protects against everything except dimension.**

Short version. The four selection principles answer four different questions. Bayes asks: optimal given prior? Admissibility asks: strictly dominated? Minimax asks: worst case? Invariance asks: respects symmetry? These can disagree (and do, in dimension three and higher), and that disagreement is structural, not a defect.

Cross-reference: which tool to reach for

If you are asked to...	**Use...**
Compute risk of a specific rule δ	Definition: $R(\theta, \delta) = \mathbb{E}_{P_\theta}[L(\theta, \delta(X))]$. For squared loss, bias–variance: $R = \text{Bias}^2 + \text{Var}$.
Find the Bayes rule for given π	**Extensive form** (Result 1, §4.4): minimize posterior expected loss for each x. For squared loss, the posterior mean.
Show a rule is admissible	Thm 8 (unique Bayes rule): show it minimizes Bayes risk uniquely. Or Thms 7 / 9 for the technical conditions.
Show a rule is inadmissible	Construct a dominating rule. For Stein-type problems, compare SURE pointwise.
Improve a non-T-based estimator	Rao–Blackwell (Thm 4): condition on the sufficient statistic T.
Find a minimax rule (proper prior)	Thm 17: find π such that the Bayes rule δ^π has *constant risk* in θ.
Find a minimax rule (improper prior)	Thm 18: find proper priors π_n with $\lim r(\pi_n) = c < \infty$, and a rule δ^* with $R(\theta, \delta^*) \leq c$ everywhere.
Prove a Stein-type dominance	Stein's identity + SURE: compute $\widehat{R}(\mathbf{x}) = p + 2\nabla \cdot \mathbf{g} + \lVert\mathbf{g}\rVert^2$ and compare pointwise.
Find a minimax rule for an invariant problem	Compute the Pitman / best equivariant rule. By Hunt-Stein (amenable groups: location, scale), this is minimax.
Find the best location-equivariant rule under squared loss	Pitman estimator: $\delta^*(X) = X_n - \mathbb{E}_0[X_n \mid T(X)]$, equivalently the generalized Bayes rule under $\pi(\theta) = 1$.
Decide whether minimizing $\mathbb{E}[L]$ is reasonable	Utility theory (Layer 2): it is forced by the rationality axioms.

Five common confusions

1. **"Which expectation?"** $\mathbb{E}_{P_\theta}$ vs. $\mathbb{E}^\pi$ vs. $\mathbb{E}^{\pi(\theta|x)}$. Ask: is θ fixed or random? Frequentist risk fixes θ. Posterior expected loss fixes x and treats θ as random. Bayes risk averages over both.
2. **"Improper prior $\Rightarrow$ no Bayes rule?"** No: you get a *generalized Bayes rule*. The posterior $\pi(\theta \mid x) \propto f(x \mid \theta)\pi(\theta)$ is computed the same way. What fails is the Bayes-risk machinery ($r(\pi, \delta) = \infty$); admissibility (Thms 7–9) and minimaxity (Thm 17) need Thm 18 as a workaround.
3. **"Why is utility in this course?"** Three uses: (a) it *justifies* $\mathbb{E}[L]$ as the right criterion; (b) it *forces* L bounded, which makes Thms 7–9 work; (c) it *provides the formal recipe* $L = -U$. Without it, the framework is ad hoc. See Layer 2 above.
4. **Sufficiency vs. Bayes vs. minimax.** Sufficiency is a property of the *data* (a statistic on $\mathcal{X}$). Bayes and minimax are *selection principles for rules*. Sufficiency reduces the search space (every rule has a replacement based on T); Bayes and minimax pick a specific rule within the reduced space.
5. **Bayes rule vs. Bayes action.** A *Bayes action* is for a fixed x: the action minimizing posterior expected loss. A *Bayes rule* is the function $\delta^\pi(x)$ returning the Bayes action for each x. Result 1 says these are equivalent: pointwise

minimization gives a rule that also minimizes the Bayes risk.

Equation reference card

Object	Formula
Frequentist risk	$R(\theta, \delta) = \mathbb{E}_{P_\theta}[L(\theta, \delta(X))]$
Bayes risk of δ	$r(\pi, \delta) = \mathbb{E}^\pi[R(\theta, \delta)] = \int R(\theta, \delta)\, dF^\pi(\theta)$
Posterior expected loss	$\rho(\pi(\theta \mid x), a) = \mathbb{E}^{\pi(\theta\|x)}[L(\theta, a)]$
Posterior	$\pi(\theta \mid x) = f(x \mid \theta)\pi(\theta)/m(x), \quad m(x) = \int f(x \mid \theta)\pi(\theta)\, d\theta$
Bayes rule (squared loss)	$\delta^\pi(x) = \mathbb{E}^{\pi(\theta\|x)}[\theta]$ (posterior mean)
Result 1 identity	$r(\pi, \delta) = \mathbb{E}^{m(x)}[\rho(\pi(\theta \mid x), \delta(x))]$
Loss from utility	$L(\theta, a) = -U(\theta, a)$ (regret: $\ell(\theta, a) = \sup_{a'} U(\theta, a') - U(\theta, a)$)
Rao–Blackwell rule	$\delta_1(t) = \mathbb{E}[\delta_0(X) \mid T = t]$
Stein's identity (Gaussian)	$\mathbb{E}_\theta[(\mathbf{X} - \theta)^T \mathbf{g}(\mathbf{X})] = \mathbb{E}_\theta[\nabla \cdot \mathbf{g}(\mathbf{X})]$
SURE	$\hat{R}(\mathbf{x}) = p + 2\nabla \cdot \mathbf{g}(\mathbf{x}) + \\|\mathbf{g}(\mathbf{x})\\|^2$ for $\delta = \mathbf{x} + \mathbf{g}(\mathbf{x})$
James–Stein estimator	$\delta^{JS}(\mathbf{x}) = (1 - (p-2)/\\|\mathbf{x}\\|^2)\mathbf{x}$
James–Stein risk ($p \geq 3$)	$R(\theta, \delta^{JS}) = p - (p-2)^2 \mathbb{E}_\theta[1/\\|\mathbf{X}\\|^2]$
Binomial minimax	$\delta(x) = (x + \sqrt{n}/2)/(n + \sqrt{n})$
Equivariance condition	$\delta(gx) = \tilde{g}\, \delta(x)$ for all $g \in \mathcal{G}$
Pitman estimator (location)	$\delta^*(X) = \int u \prod_i f(X_i - u)\, du \,/\, \int \prod_i f(X_i - u)\, du$

What we did not cover

Berger Chapters 7 and 8 are natural next reading, but we have run out of time.

- **Sequential analysis (Ch 7).** Optimal stopping. The decision becomes *when to collect more data, vs. when to act.* The SPRT (sequential probability ratio test) is the minimax sequential rule. Builds directly on the Bayes machinery of Layer 4.
- **Complete classes (Ch 8).** Under regularity conditions, the admissible rules are essentially the Bayes rules. This is the deep structural result that closes the loop between admissibility (Layer 5) and Bayes machinery (Layer 4): admissibility and Bayes-ness are nearly equivalent.

I will still post the slides for these two topics for your reference, but they will not be covered in Midterm 2. Midterm 2 covers Lectures 6 through 11: minimax, statistical games, Stein's phenomenon, and invariance.